MODULE 14

ATM and Card Crime Investigation

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

The criminal doesn't steal plastic - **he steals information**. This module follows that information.

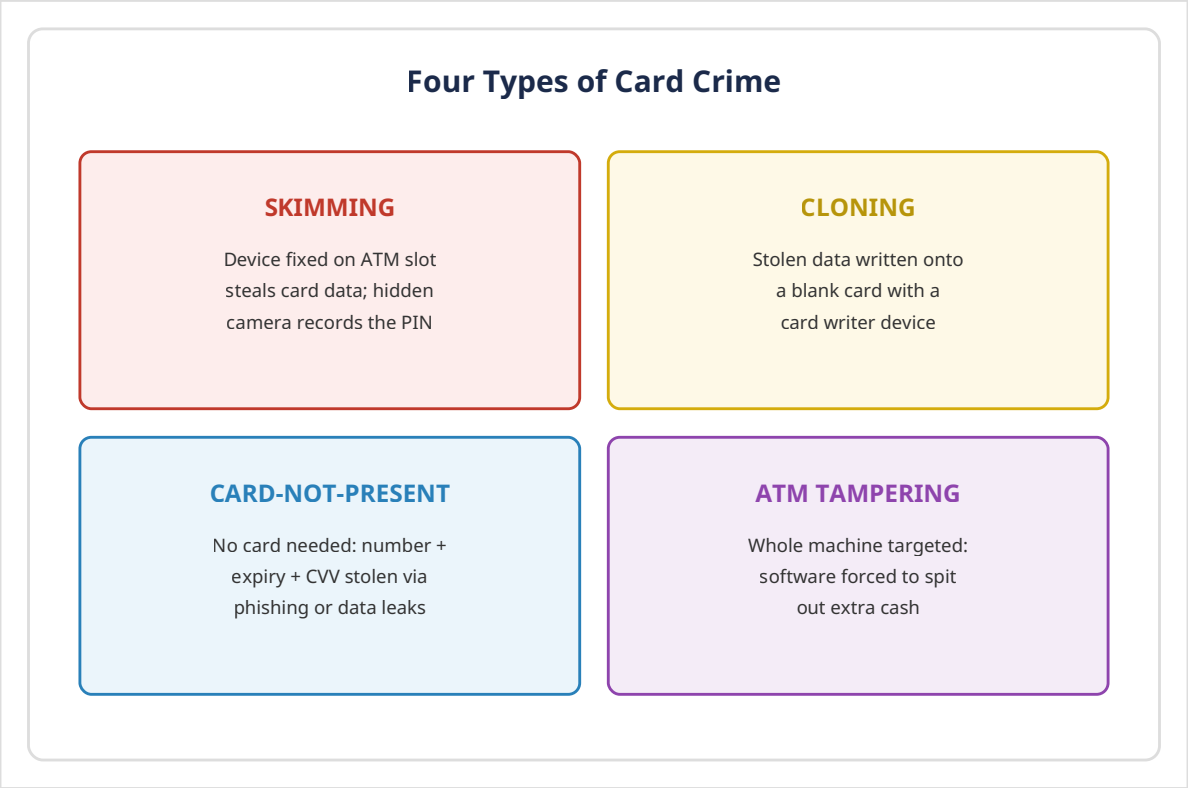
- **Four card-crime types** - skimming, cloning, card-not-present, ATM tampering
- **Spotting skimmers** - ten signs, and how to secure one as evidence
- **Transaction anomalies** - impossible travel and velocity in bank logs
- **NFC and card writers** - what they reveal, what they don't
- **Victim questioning** - the four question groups that set direction
- **Legal sections** - IT Act 66C/66D, BNS 318/319, applied correctly

The ATM That Watched Your Pocket

You withdraw cash. Everything looks normal. But a **thin fake cover** on the card slot copied your card:

- A hidden camera above the keypad **recorded your PIN** through a pinhole
- Days later, money leaves your account from a city you never visited
- This is **skimming** - and it is only one of four card-crime types

Four Types of Card Crime



- Skimming feeds cloning: **stolen data + card writer = fake card**
- Card-not-present needs no device at all - just leaked numbers
- Know the type first - **each type has a different investigation path**

Investigation: Five Fixed Steps



- Step 3 is time-critical: ATM footage **auto-deletes** - secure it on day one
- Step 4 finds the skimmer: many victims, one area, one week = **inspect that ATM**

Impossible Travel: Fraud Confesses Itself



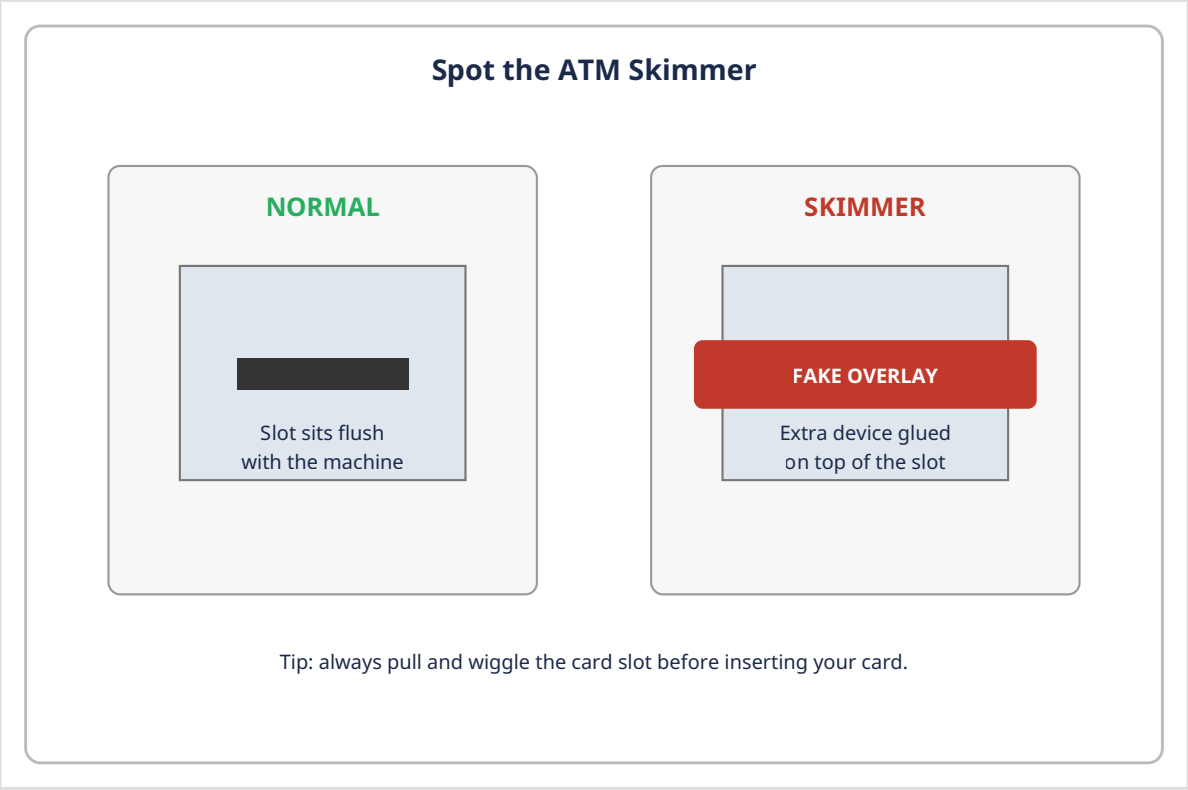
- Delhi 09:17, Chennai 09:31 - **14 minutes, 1,700 km**. No human travels that fast
- Always confirm with the holder: 'were you outside Delhi?' - then it becomes **evidence**

Velocity: Too Fast to Be Human

Six withdrawals of Rs 10,000 in **15 minutes** - no genuine customer behaves like this:

- Count per card, per window: frequency + amount = the **velocity score**
- Criminals rush because the card may be **blocked any minute**
- Normal pattern for contrast: same card, hours apart, small amounts - **learn both**

Spot the ATM Skimmer



- Compare with the bank's **other machines** - uniformity is the baseline
- Loose slot + glue marks + odd thickness = **strong combined signal**
- Suspect it? Don't touch - **photograph from distance, call the bank**

Stealing the PIN: Two Methods

Card data is half the crime. The **PIN** is the other half - stolen two ways:

- **Hidden camera:** pinhole in the panel above the keypad, eye on your fingers
- **Fake keypad:** thin overlay pasted on the real one - feels raised, sounds different
- Public advice you give: **cover the keypad with your other hand** - defeats both

Securing a Skimmer as Evidence

A recovered skimmer can carry **fingerprints** - handle it like the treasure it is:

- **Gloves on** before touching anything - your prints must not join the criminal's
- Photograph **in place first**: overall, sides, close-ups with scale
- Seal in a separate bag, tag it: case number, date-time, seizer's signature - **chain of custody starts here**

NFC: The Tap-to-Pay Question

'My card never left me!' - then check the **contactless angle**:

- NFC reads card signals from centimetres away - in a crowd, unnoticed
- But: **dynamic cryptogram** changes every transaction - one read can't be replayed
- Banks cap contactless amounts and verify with servers - **full cloning via NFC is hard**
- Investigation checklist: contactless transaction? within limit? holder's location? valid cryptogram?

Card Writers: Raid Gold

In a gang raid, a **card writer + blank cards + stolen data files** = the case:

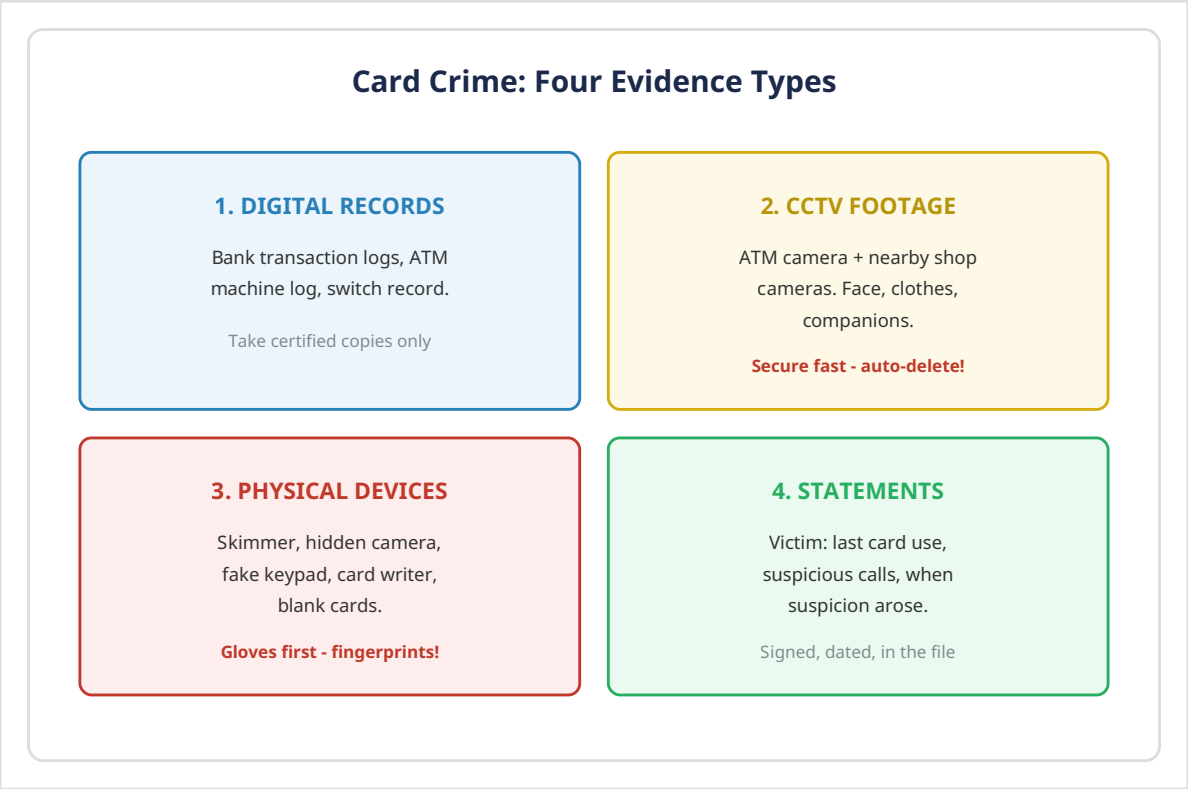
- The device's memory reveals **how many cards** were written - scale of the crime
- Links the gang to every cloned card in circulation
- Same warning as skimmers: **knowing is for investigation**; making, buying or using is a serious crime

Card-Not-Present: Theft Without the Card

No skimmer, no ATM - just **number + expiry + CVV**, stolen online:

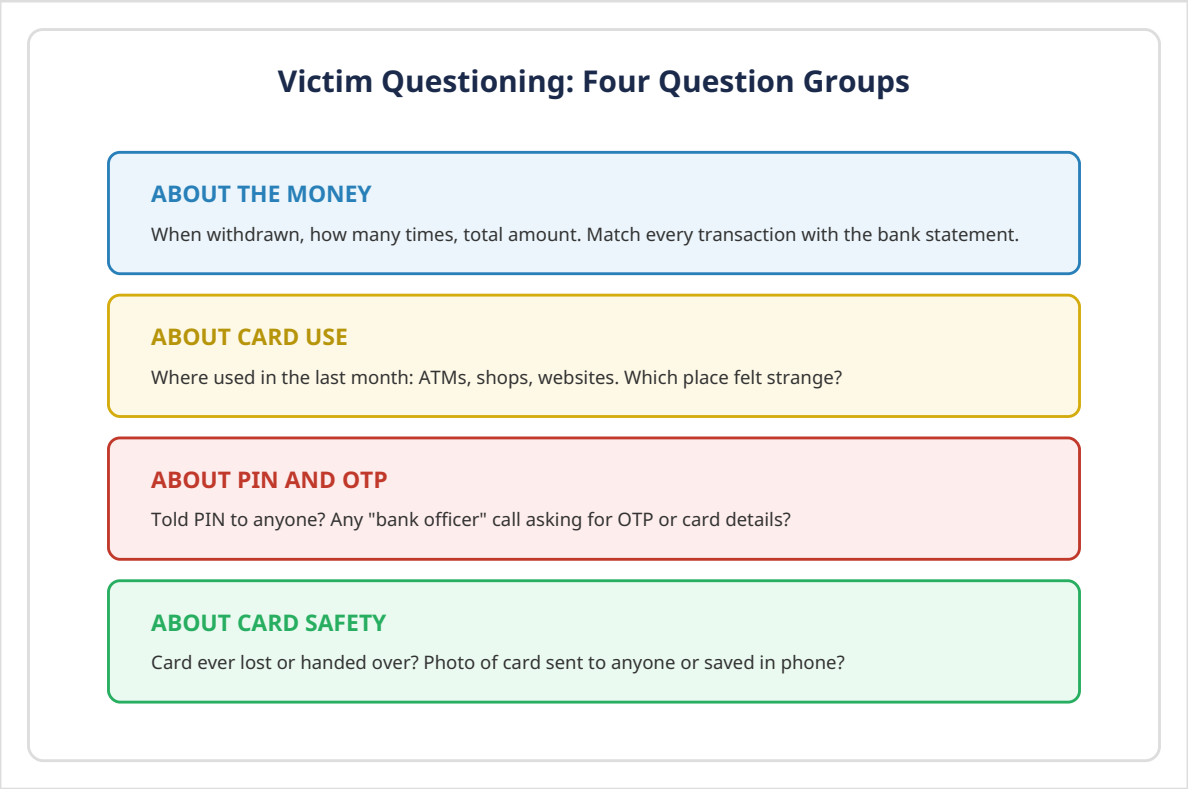
- Leak sources: phishing sites, data breaches, malicious apps - **ask the victim about all three**
- The **payment gateway's** logs show: which site, which device, which IP paid
- The **delivery address** is the classic mistake - goods must go somewhere; gangs get caught here

Four Evidence Types, One Timeline



- Join all four into **one timeline**: skimmer fixed → data stolen → card made → cash out
- Certified copies of logs - **uncertified logs weaken in court**

Questioning the Victim: Four Groups



- First questioning **sets the investigation's direction** - get it right
- Write answers on a form, take the **victim's signature** - it joins the case file

Legal Sections at a Glance

Section	Covers	Used when
IT Act 66C	Identity theft	Skimming, cloning card data
IT Act 66D	Cheating by computer	Online card fraud, phishing
BNS 318	Cheating	Every fraud with loss
BNS 319	Cheating by personation	Posing as bank officer

Real Case: The Cosmos Bank Heist, 2018

Pune's Cosmos Bank - **Rs 94 crore** vanished in hours, across 28 countries:

- Attackers breached the bank's **ATM switch** - the system behind every ATM transaction
- Fake cards withdrew cash **simultaneously worldwide** - faster than the bank could react
- Transaction logs were the weapon: country-wise ATM trails joined into one conspiracy
- Source The Indian Express

Real Case: The 2016 Debit Card Leak

32 lakh debit cards endangered - India's biggest card-security scare:

- Malware inside a payment company's system **silently harvested** card data for months
- Banks blocked lakhs of cards overnight; customers rushed to change PINs
- Lesson: card data is stolen **anywhere in the payment path** - not just at ATMs
- Source The Economic Times

Your Turn: Labs in This Module

Four labs - **observe, never tamper**:

- **Lab 1:** 10-point skimmer checklist in your own words + ATM observation
- **Lab 2:** NFC read of your OWN card - what shows, what doesn't, and why
- **Lab 3:** Forensic photography drill - 10+ photos, custody tag, photo log
- **Lab 4:** Find impossible travel + velocity in sample logs - write the 3-line conclusion

Mistakes That Weaken Card Cases

Card cases are won on **procedure**. These errors lose them:

- Touching a skimmer **bare-handed** - your fingerprints join the evidence
- Waiting a week for CCTV - **auto-delete** has already eaten it
- Using **uncertified** bank logs - the defence will tear them apart
- One weak signal = conclusion - **combinations of strong signals** decide

First Response Checklist

Card fraud reported? **Do this today, not tomorrow:**

- **1930 + bank:** block the card, flag the transactions - stop the bleeding
- **CCTV:** secure ATM footage in writing - before auto-delete
- **Logs:** certified transaction log from the bank - which ATM, what time
- **Victim form:** four question groups, signed - direction for the whole case

Key Takeaways

If you remember nothing else from Module 14:

- Criminals steal **information, not plastic** - follow the information
- **Impossible travel + velocity** in logs = the fraud confessing
- Skimmer evidence: **gloves, photograph in place, seal, tag** - fingerprints matter
- CCTV is **perishable** - secure it on day one or lose it forever
- Question the victim in **four groups**; get it signed
- Right sections: **66C/66D IT Act, 318/319 BNS** - with senior guidance

Review Questions

- 1. How does skimming differ from card-not-present fraud? Where does the criminal get the data in each?
- 2. List five skimmer signs from Lab 1 and rank them strong vs weak signals.
- 3. Card ...4521: Delhi 09:17, Chennai 09:31. Walk through your confirmation steps before calling it fraud.
- 4. Why is a card writer found in a raid such powerful evidence? Name two handling precautions.
- 5. A victim's card never left her purse but money was withdrawn. List your four NFC-angle checks.

NEXT: MODULE 15

Wallet and QR Code Fraud Investigation

UPI, mule accounts, and the money trail